Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/17/2004
PI Name:	Knobloch, Edgar Ph.D.		
Project Title:	Structure and Dynamics of Free surface flows with heating and vibration		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-7300	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	11/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Vega, Jose Manuel (ETSI Aeronauticos) Martel, Carlos (ETSI Aeronauticos) Thiele, Uwe (Max Planck Intitute for Complex Systems)		
Grant/Contract No.:	NNC04GA47G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project seeks to understand the effects of vibration on fluid systems. Two systems are of primary interest, the so-called Faraday system, in which a container of liquid is subjected to vertical acceleration, and the properties of drops and thin liquid films on a vibrating substrate. The focus of the project is on fundamental fluid dynamics involving free surface flows, contact line dynamics and the effects of heating and/or spatial inhomogeneities of the substrate. The objective to obtain a broad understanding of the effects of vibration on pattern-forming instabilities arising in thin liquid layers as a result of Marangoni stress, wetting phenomena and contact line dynamics. These processes are of fundamental importance in the flow of liquids under microgravity conditions. Under these conditions Marangoni effects can be used to promote lubrication, coating and/or cooling but these processes are associated with instabilities that may be reduced or suppressed (and in some cases enhanced) by residual g-jitter.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The understanding achieved will be of fundamental importance for related phenomena on the earth, in particular in the currently exciting area of nanofluidics.
Task Progress:	In the year 1.11.03-30.10.04 the PI has pursued two main themes: (i) The dynamics of the Faraday system, i.e., the properties of surface gravity-capillary waves in vertically vibrating container of liquid. When the amplitude of vibration exceeds a critical value the flat surface becomes unstable to standing waves, which in turn are subject to a variety of secondary instabilities at finite amplitude. In low viscosity liquids the most important of these are associated with the excitation of a streaming flow driven through Reynolds stresses generated in oscillatory viscous boundary layers at rigid walls and the free surface. In this regime the surfaces waves are described by a set of coupled amplitude-streaming flow equations. This system constitutes a new class of pattern-forming equations. (ii) The structure and stability properties of thin liquid films on a heated vibrating horizontal substrate, and the dynamics of such films flowing down a heated vibrating plate. In the following more details of the work carried out are given: (i) The PI has studied the dynamics described by the coupled amplitude-streaming flow equations derived earlier for a liquid layer in an elliptically distorted right circular cylinder. In general this system consists of a pair of equations for the amplitude of clockwise and counterclockwise traveling waves, coupled to a Navier-Stokes-like equation for the streaming flow. The streaming flow is driven through boundary conditions imposed at the undisturbed surface, obtained by matching the boundary layer solution to the flow in the bulk. Similar (but different) boundary layers are present at the container walls. In nearly inviscid liquids the terms describing this coupling to the streaming flow are as important as the intrinsic nonlinearities; the coupling must therefore be included. The PI has found that the excitation of the streaming flow is associated with instabilities that lead to different types of drift of the standing waves that appear at onset of the instability. To study these the PI uses the amplitude equation coefficients derived by Miles, and uses the eccentricity of the cylinder to move secondary symmetry-breaking instabilities towards small amplitude where they can be studied using a simplified system of equations. This procedure led to the discovery of unexpected relaxation oscillations in this system, whose origin was traced to the disparity in time scales between the (slow) decay time of free surface gravity-capillary waves, and the (even slower) decay time of the streaming flow, that is characteristic of low viscosity liquids. These relaxation oscillations are important because they are often the first state that is observed as the amplitude of the parametric forcing increases and the flat surface loses stability. In addition the PI has observed abrupt changes in the amplitude of these oscillations associated with the so-called canard phenomenon, as well as delayed loss of stability that arises when the system drifts slowly through a Hopf bifurcation. Many of these discoveries resemble behavior exhibited in models of bursting biological systems, but this is the first time they have been seen in a fluid system. An extended review of burst processes in fluid dynamics by the PI has recently appeared; this review puts the phenomena mentioned above into a broader context. Publications: M. Higuera, E. Knobloch and J.M. Vega. Relaxation oscillations in a nearly inviscid Faraday system. Theoret. Comput. Fluid Dyn. 18, 323-333 (2004). E. Knobloch. A finite-dimensional mechanism responsible for bursts in fluid mechanics. In: Dynamics and Bifurcation of Patterns in Dissipative Systems, G. Dangelmayr and I. Oprea (eds.), World Scientific Series on Nonlinear Science, Vol. 12, World Scientific, Singapore, pp. 231-273 (2004). (ii) The PI has derived a fully nonlinear amplitude equation for long wavelength perturbations of the free surface of a thin liquid film a function of the thin film parameters: the Biot number, the Marangoni number, the Galileo number and a new parameter, the vibration number, as well as the substrate inclination. This equation is valid in both one and two dimensions. Properties of drops and thin films described by this equation in one dimension have been examined. The findings include a description of how the shape of a drop on a heated plate and the internal circulation within it change as a result of vibration, and how vibration speeds up the flow of a thin film down an inclined (heated) plate. This work, jointly with the coIs U Thiele and JM Vega, is about to be submitted for publication in the Journal of Fluid Mechanics. This work is now being extended to two-dimensional films. In addition the effect of pinning of drops due to surface microstructure is being studied, with a view to elucidating the process by which drops begin to slide downslope. An article on this system in the absence of vibration has appeared (see below). Publications: U. Thiele and E. Knobloch. Thin liquid films on a slightly inclined heated plate. Physica D 190, 213-248 (2004).
Bibliography Type:	Description: (Last Updated: 07/20/2021)
Articles in Peer-reviewed Journals	Thiele U, Knobloch E. "Thin liquid films on a slightly inclined heated plate." Physica D: Nonlinear Phenomena. 2004 Apr;190(3-4):213-48. https://doi.org/10.1016/j.physd.2003.09.048 , Apr-2004
Articles in Peer-reviewed Journals	Higuera M, Knobloch E, Vega JM. "Relaxation oscillations in a nearly inviscid Faraday system." Theor Comput Fluid Dyn. 2004 Nov;18:323-33. https://doi.org/10.1007/s00162-004-0144-2 , Nov-2004

Books/Book Chapters	Knobloch E. "A finite-dimensional mechanism responsible for bursts in fluid mechanics." in "Dynamics and Bifurcation of Patterns in Dissipative Systems." Ed. G. Dangelmayr, I. Oprea. New Jersey: World Scientific, 2004. p. 271-273. https://doi.org/10.1142/9789812567840_0013 , Nov-2004